1 Notation and conventions

Indices and sets. Indices are lower-case Latin letters; sets are calligraphic upper-case:

- $r \in \mathcal{R}$ is the receiver. Single-receiver modes (PPP, PPP-RTK) have $|\mathcal{R}| = 1$. Baseline modes have a fixed reference and one or more rovers. The network estimator (`PPP_RTK_Network`) processes a peer set of $|\mathcal{R}| \geq 2$ stations (50–305 in practice) with no reference/rover asymmetry.
- $g \in \mathcal{G} = \{\text{GPS}, \text{GAL}, \text{BDS}, \text{GLO}, \text{QZS}, \dots\}$ is the constellation.
- $i \in \mathcal{S}_g$ is a satellite of constellation g. p_g denotes the pivot satellite of constellation g under the BSSD parameterisation (Section 4).
- $s \in \mathcal{B}$ is the signal/observation type, identified by its RINEX-3 three-character code (e.g. `C1C`, `L2W`, `L5X`).
- $k \in \mathbb{N}$ is the discrete epoch index.

Vectors and matrices. Vectors are bold lower-case ($\boldsymbol{x}$, $\boldsymbol{r}$); matrices are bold upper-case ($\boldsymbol{H}$, $\boldsymbol{P}$). $\boldsymbol{a}^{\mathrm{T}}$ is the transpose. A hat denotes a state estimate and a tilde a one-step prediction: $\hat{\boldsymbol{x}}_{k|k}$ and $\tilde{\boldsymbol{x}}_{k|k-1}$.

Observables. Each receiver/satellite/signal triple supplies two real-valued observables per epoch, both in metres:

- $P^i_{r,s}$ is the pseudorange (code), in metres.
- $\Phi^i_{r,s}$ is the carrier phase, scaled to metres by λ_s. In addition to the range, clock and atmosphere terms it shares with the code, it carries an ambiguous (non-range) part: $\lambda_s N^i_{r,s}$ plus the fractional phase biases $b_{r,s} - b^i_s$ (Equation (5)).

Where the distinction matters, s also denotes the band ($L_1, L_2, L_5, \dots$) and the tracking attribute (e.g. `C`, `W`, `X`). The leading observable-type letter (`C`/`L`) selects code or phase. The frequency-only quantities f_s, λ_s and $N^i_{r,s}$ depend only on the (band, attribute) part, so a code signal and its sibling phase share them. The state keys follow the same split: ambiguity keys carry the full phase code (`"G01_L1C"`), receiver code-bias keys only the band digit (`"GPS_IFB2"`).

Frequencies and wavelengths. f_s is the carrier frequency (Hz) and $\lambda_s = c/f_s$ the wavelength (m). The frequency-squared factor that maps the L1-equivalent slant ionospheric delay onto signal s is

$$\mu_s \;:=\; \frac{f_1^2}{f_s^2}. \tag{1}$$

Here f_1 is the GPS L1 frequency (1575.42 MHz), the common reference for all constellations; it is not the band-1 frequency of the satellite's own system. The estimated state $I^i_{r,1}$ is therefore GPS-L1-equivalent for BDS, GLO and GAL satellites as well (the slant-ionosphere term of `measurement/ObservationEquation.cpp`, `addIonosphere`: `f_ref = Constants::GPS_L1_FREQ`).

Differencing operators. Three differencing operators recur throughout:

$$\Delta q^{i,p_g}_{r,s} := q^i_{r,s} - q^{p_g}_{r,s}, \quad \nabla q^i_{r_1 r_2,s} := q^i_{r_1,s} - q^i_{r_2,s}, \quad \nabla\Delta q^{i,p_g}_{r_1 r_2,s} := \Delta\big(\nabla q^i_{r_1 r_2,s}\big).$$

Δq is the between-satellite single difference of q relative to the pivot p_g, ∇q the between-receiver single difference, and $\nabla\Delta q$ the double difference.

Sign conventions. The slant ionospheric delay $I^i_{r,s}$ enters the code observable with a positive sign and the carrier phase observable with a negative sign; this sign difference produces code–phase divergence. All other range-additive terms (geometry, clocks, troposphere, biases) carry the same sign on both observables. Equations (4) and (5) give the explicit equations. In the implementation the satellite bias parameters (code and phase) enter each row with the partial $+1$ and the satellite clock correction with the partial -1, one constant each shared by every code that builds the row (Section 3.4); a bias parameter therefore carries the sign of the term as it appears in the observation, $-B^i_s$ or $-b^i_s$ of Equations (4) and (5).

Symbol table.

Symbol	Units	Meaning
$\boldsymbol{r}_r, \boldsymbol{r}^i$	m	ECEF position of receiver/satellite
ρ^i_r	m	geometric range $\lVert\boldsymbol{r}^i - \boldsymbol{r}_r\rVert$
$\boldsymbol{e}^i_r$	none	unit LOS from r to i in ECEF
$c\,\mathrm{d}t_r$	m	receiver clock offset (scaled by c)
$c\,\mathrm{d}t^i$	m	satellite clock offset (CLK product in user modes; the estimated correction `"<sat>|SC"` in `PPP_RTK_Network`)
$c\,\delta t^{(g)}_r$	m	receiver inter-system time offset for constellation $g \neq g_{\mathrm{ref}}$
T^i_r	m	total tropospheric path delay
$\mathrm{ZHD}^i_r, \mathrm{ZWD}^i_r$	m	zenith hydrostatic / wet components
$I^i_{r,1}$	m	L1-equivalent slant ionospheric delay
$N^i_{r,s}$	cyc	integer carrier-phase ambiguity
$b_{r,s}, b^i_s$	m	receiver / satellite phase biases
$\bar{b}^i_s$	m	estimable satellite phase bias (network mode; Section 2.5)
$B_{r,s}, B^i_s$	m	receiver / satellite code biases (OSB)
$B^{(g,b)}_{r,k}$	m	per-FCN-slot GLONASS receiver IFB, band b; here $k \in [-7,+6]$ is the signed FDMA channel number (a context-local reuse, not the epoch index)
$\varepsilon^P, \varepsilon^\Phi$	m	residual (multipath + thermal)
$\boldsymbol{v}_r$	m/s	receiver velocity (optional / IMU state)
$\lambda_{\mathrm{WL}},\ \lambda_{\mathrm{NL}},\ \lambda_{\mathrm{EWL}}$	m	wide- / narrow- / extra-wide-lane wavelength $c/(f_a \mp f_b)$ for the lane's band pair
λ_{IF}	m	effective ionosphere-free wavelength $\lvert c_1\lambda_1 - c_2\lambda_2\rvert$ with the IF coefficients $c_1 - c_2 = 1$; not an integer wavelength (the cycle-slip document)
$\boldsymbol{C}^e_b$	none	body-to-ECEF attitude (direction-cosine) matrix

2 State vector layout

The Kalman filter operates on a state vector $\boldsymbol{x}_k \in \mathbb{R}^{n_k}$ whose dimension n_k varies across epochs. A per-satellite or per-(satellite, signal) state is appended when its satellite or signal is first required, and removed when it is absent from the epoch's measurement set (for example, after the satellite sets below the elevation mask). A cycle slip does not remove the ambiguity state; it re-seeds it (see the ambiguity paragraph below). `StateManager` holds all bookkeeping: the key→index maps, the appending of rows, and the compaction of $\boldsymbol{x}$ and $\boldsymbol{P}$ on removal, which shifts every later index down by one (there is no free list). `KalmanFilter` reads the state through that interface and never indexes it by hard-coded offsets. The dynamic part of the vector is a set of families, each with one key→index map, declared once in a table (Section 2.3).

This section describes the single-receiver layout (PPP, PPP-RTK and the baseline modes). The network estimator keeps no `StateManager`: each of its stations owns a parameter set of its own, described in Section 2.5.

2.1 Schematic layout

In schematic form, suppressing the epoch index k:

$$\begin{aligned}\boldsymbol{x} \;=\; \big[\; & \underbrace{\boldsymbol{r}_r^{\mathrm{T}}}_{3},\ \underbrace{\boldsymbol{v}_r^{\mathrm{T}}}_{0\text{ or }3},\ \underbrace{c\,\mathrm{d}t_r}_{1},\ \underbrace{(c\,\delta t_r^{(g)})^{\mathrm{T}}_{g\neq g_{\mathrm{ref}}}}_{|\mathcal{G}|-1},\\ & \underbrace{\mathrm{ZWD}_r,\ (g_N, g_E)}_{1+0\text{ or }1+2},\ \underbrace{(c\,\mathrm{d}t_r^{\phi})}_{0\text{ or }1},\\ & \underbrace{(I_{r,1}^i)^{\mathrm{T}}_{i\in\mathcal{V}_k}}_{|\mathcal{V}_k|},\ \underbrace{(N_{r,s}^i)^{\mathrm{T}}_{(i,s)\in\mathcal{A}_k}}_{|\mathcal{A}_k|},\\ & \underbrace{(B_{r,s})^{\mathrm{T}}_{s\in\mathcal{B}}}_{|\mathcal{B}|},\ \underbrace{(B_s^i)^{\mathrm{T}}_{(i,s)\in\mathcal{D}_k}}_{|\mathcal{D}_k|}\;\big]^{\mathrm{T}}. \end{aligned} \tag{2}$$

where

- $\mathcal{V}_k$ is the set of satellites visible at epoch k (above the elevation mask, with valid orbit and clock products and at least one tracked signal);
- $\mathcal{A}_k \subseteq \mathcal{V}_k \times \mathcal{B}$ is the set of phase-tracked (satellite, signal) pairs;
- $\mathcal{D}_k \subseteq \mathcal{V}_k \times \mathcal{B}$ is the subset of (satellite, signal) pairs that carry a satellite code-bias state: the pairs for which the BIA product lacks a satellite OSB (`estimation.estimate_sat_biases`).

The schematic omits several conditional refinements, detailed below: the rate states that accompany velocity (a receiver clock rate and one inter-system clock-rate offset per non-reference constellation, both absent under BSSD); the optional per-satellite slant-ionosphere rate under `enable_iono_rate`; the receiver inter-signal code biases; the nine-state inertial augmentation of the inertial-aiding document; and the removal of the whole receiver-side clock and inter-system-offset block under the BSSD parameterisation (Section 4.1).

The schematic groups the states by physical role; it is not the storage order. `StateManager` places a fixed core first (`StateManagerLayout.cpp`), in the order position, receiver clock (absent under BSSD), ZWD, velocity, receiver clock rate, phase clock, the two troposphere gradients, the nine inertial states, the inter-system time offsets and the inter-system clock-rate offsets, each present only when enabled. Every dynamic state (ionosphere, ionosphere rate, ambiguity, receiver code bias,

satellite code bias) is appended after the core in order of creation, so the categories interleave; each category is addressed only through its key map, as described under dynamic indexing below.

2.2 Component-by-component description

Position $\boldsymbol{r}_r \in \mathbb{R}^3$. ECEF Cartesian position of the antenna reference point. Always present. `Application::initializePosition` seeds it from the RINEX header approximate position, shifted from the marker to the antenna reference point by the header's antenna H/E/N offset (or from the Station Information File coordinate when that a-priori source is enabled). The seed must lie within the linearisation radius of the geometric range model (Section 3.3). The per-epoch Weighted Least-Squares (WLS) solution seeds the receiver clock (on the first epoch, after a clock jump and after a data gap); it re-seeds the position only in the kinematic gap-recovery branch of `Application::handleGap`.

Velocity $\boldsymbol{v}_r \in \mathbb{R}^3$. Optional. Enabled by `estimation.estimate_velocity = true`; tightly-coupled IMU integration forces it on. With velocity present, position and velocity form a coupled continuous-white-noise-acceleration (CWNA, integrated-random-walk) block driven by a single acceleration spectral density, `process_noise.acceleration_psd_q` (the CWNA process-noise model in the estimation document); the scalar `position_q` of the velocity-free case is then unused. When BSSD is off, velocity estimation also adds a receiver clock-rate state $c\,\dot{\mathrm{d}}t_r$ and one inter-system clock-rate offset (ISBR) per non-reference constellation, each CWNA-coupled to its bias counterpart (`clock_rate_q`, `system_time_rate_offset_q`). Under BSSD both rate states are omitted, because same-system single differencing cancels them.

Receiver clock $c\,\mathrm{d}t_r$. Single state, in metres. Its process noise is large (`clock_bias_q` $\sim 10^4\,\mathrm{m}^2$ per second), so the state is in effect re-estimated each epoch. This follows oscillator drift without imposing a stochastic model that may not match the receiver's oscillator. Under the BSSD (between-satellite single-difference) parameterisation (Section 4.1) the receiver clock is unobservable and the state is not allocated; neither are the inter-system offsets below, so PPP-RTK SD mode has no receiver-side clock block.

Inter-system time offsets $c\,\delta t_r^{(g)}$. One per active constellation other than GPS, which is the fixed reference in `StateManager`. These absorb the per-constellation clock biases that arise from mixing analysis-centre clock products across systems ("GAL − GPS time offset," etc.).

Tropospheric ZWD and gradients. ZWD_r is the zenith wet delay. The zenith hydrostatic delay ZHD_r is deterministic, from the Saastamoinen model [4]. The total slant tropospheric delay is

$$T_r^i \;=\; m_h(\theta_r^i)\,\mathrm{ZHD}_r \;+\; m_w(\theta_r^i)\,\mathrm{ZWD}_r \;+\; m_g(\theta_r^i)\big[g_N\cos\alpha_r^i + g_E\sin\alpha_r^i\big], \tag{3}$$

with elevation θ_r^i, azimuth α_r^i, the mapping functions m_h, m_w (the Niell tables by default, labelled `Gmf` in the configuration, or VMF3 with GPT3 coefficients when configured) and the Chen–Herring gradient mapping $m_g(\theta) = 1/(\sin\theta\tan\theta + 0.0032)$ [1] (the atmosphere document). The gradients (g_N, g_E) are optional.

Per-satellite slant ionosphere $I_{r,1}^i$. Uncombined processing in every mode except the baseline modes (`processing.uncombined_enabled`, forced on for the PPP-RTK family; `GnssConfig::`

carriesSlantIonosphere). One state per visible satellite, expressed at the L1 frequency; signal-specific delays follow Equation (1). The seed is formed from the geometry-free combination of the satellite's BIA-covered code residuals (seedIonosphere, the first-epoch seeding section of the estimator-infrastructure document). With fewer than two covered codes on different bands, the seed is zero and the initial standard deviation is inflated tenfold. The initial variance, like the process noise, is scaled by the obliquity factor. The process noise is a random walk whose zenith spectral density ionosphere_q is projected onto the slant by the squared obliquity factor $1/\sin^2\theta_r^i$ (the ionospheric process-noise model in the estimation document), so low-elevation states vary faster. It is applied on every run that carries the states, in PPP as in PPP-RTK; without it the state would be a constant that cannot follow a slant delay moving by metres per pass. These explicit per-satellite states let PPP-RTK keep integer ambiguity recovery without forming the ionosphere-free combination, which amplifies code noise by a factor of ~ 3. Under estimation.enable_iono_rate (default off), each visible satellite also carries a slant-ionosphere rate, so that the pair forms a constant-velocity block $[\, I_{r,1}^i,\, \dot{I}_{r,1}^i \,]$. The rate $\dot{I}_{r,1}^i$ has no direct observation ($H = 0$); only the F-coupling into the level state drives it. It has its own spectral density process_noise.iono_rate_psd and its own slip response (reinflateIonosphereRateVariance), and lets the filter track a linear slant ramp rather than a pure random walk. The rate state is a satellite-specific dynamic row, not a core state: it is appended with its level state, pruned with it and treated with the other per-satellite states in gap recovery, but after prunes and re-adds it need not sit next to the level state, so it is addressed only through its own map (iono_rate_indices_).

Carrier-phase ambiguities $N_{r,s}^i$. One state per phase-tracked (i, s) pair, in metres (the state value is λN; the nl_wl basis below is the exception, carrying integers in cycles). The ambiguities are float through the prediction and update cycle; when AR is enabled, the LAMBDA-family algorithms in filter/AmbiguityResolver [6] resolve them to integers afterwards. A cycle slip calls StateManager::resetAmbiguity, which decorrelates the state rather than repairing it: the ambiguity's covariance row and column are zeroed, its diagonal is reset to σ_m^2, and the key is marked pending re-seed. The state value is left unchanged by the reset; ensureRequiredStates (in MeasurementModel) then overwrites it with a geometry-based seed (seedAmbiguity, the first-epoch seeding section of the estimator-infrastructure document). This is the project's reset-over-repair convention: a slipped ambiguity has no trustworthy correlation with the rest of the state, so its cross-covariances are cleared. A reset for a key that is not in the state (for example, a signal excluded by the band filter) is refused and creates no state.

Receiver code biases $B_{r,s}$. These absorb the receiver-side inter-frequency biases that the OSB convention does not reference to zero. They are not one per signal. The receiver-bias rule (ReceiverBiasResolver::resolveReceiverBias, measurement/ReceiverBias.hpp) keys them per (system, frequency band): both bands of the system's clock-reference signal pair map to one per-system differential code bias ("<system>_DCB"), each band with its own coefficient, and every other band gets its own inter-frequency bias ("<system>_IFB<band>"). Like the receiver clock they are receiver-common, so under BSSD they receive no process noise and the SD transform zeroes their columns. The same rule serves the network estimator, whose stations additionally give a second code of a band, one that is not the station's latched canonical code on that band, a receiver inter-signal code bias ("<system>_ISC_<signal>", unit coefficient, GLONASS excluded; network.second_code_channel or processing.channel_mode:all). The network treatment, with its carry, is in Section 2.5.

Per-slot GLONASS inter-frequency bias $B_{r,k}^{(g,b)}$. Optional, off by default. GLONASS FDMA assigns each slot $k \in [-7, +6]$ a distinct carrier frequency $f = f_{\text{base}} + k\,\Delta f$, so the receiver group delay differs per slot and a single per-band IFB state cannot represent it. With `estimation.glonass_ifb.enable = true` and its prerequisite `estimation.estimate_ifb` (without which the loader warns and the flag is inert), the receiver-bias rule instead creates one state per (FDMA band $b \in \{\text{G1}, \text{G2}\}$, slot k) with unit measurement coefficient, keyed `"GLONASS_IFB<band>_k<±slot>"`. The central channel $k = 0$ is the held datum and has no state; the GLONASS inter-system time offset absorbs its constant. The CDMA bands (G3/G4/G6) have a slot-independent frequency; with the flag on, the GLONASS branch of the rule returns no receiver bias key for them (it handles every GLONASS signal and does not fall through to the per-band path). With the flag off, GLONASS carries no per-band IFB state on a non-reference band. Unlike the receiver-common biases, these columns survive between-satellite single differencing, because two GLONASS satellites on different slots do not cancel. `BssdTransform` and `StateManager` therefore exempt them from the receiver-block zeroing and from the process-noise suppression (via `isGlonassPerSlotIfbKey`); only the inter-slot differences are estimable in SD.

Satellite code biases B_s^i. Per (satellite, signal) pair, but only for a pair whose OSB the BIA file does not supply, typically a third-frequency signal in a triple-frequency setup (`estimation.estimate_sat_biases`). The network estimator estimates its own satellite code biases as parameters instead, covering every raw code and not using the external `.BIA` (Section 2.5). These states are weakly stochastic (`sat_bias_q` $\sim 10^{-10}$), because the underlying physical biases are nearly stationary over a day.

Inertial augmentation. With IMU integration enabled, the core state gains nine entries: a three-axis attitude error and the three-axis accelerometer and gyroscope biases. The strapdown mechanisation propagates them, rather than the random-walk process model. The inertial-aiding document treats them in full.

Baseline differential troposphere. In baseline mode the core is the position alone; double differencing cancels the receiver clocks and inter-system offsets, and on a short baseline the ionosphere and troposphere, so none of those states exists. An optional single differential zenith-wet-delay state $\Delta\text{ZWD}_{r_{\text{rov}} r_{\text{ref}}}$ follows the position (`baseline.estimate_differential_tropo`, Section 4.3). This is the only mode in which `StateManager` reads its `core_state_size` constructor argument; the PPP and PPP-RTK paths ignore it and derive the core layout from the configuration.

2.3 Declared state families

The dynamic families are declared once, in `include/filter/StateFamily.hpp` (`kStateFamilies`). Each row gives the family's label prefix and snapshot tag, the variance re-inflation it answers to, its gap response, the rule that names the satellite of a key (a state whose satellite is a frozen pivot gets no process noise), its process-noise rule and spectral-density key, the configuration that enables its birth and the source of its initial variance. Every place that visits all families (the state size, the index shift on removal, the state labels, the real-time snapshot) reads the table. Table 1 lists the rows. At run start `StateManager::logFamiliesAsArmed` prints one line per family: whether the configuration births it, the label and tag, the initial standard deviation with its source, the process-noise rule with its value and the events wired to it (re-inflation, reset, prune, gap decoupling, the frozen-pivot skip, the snapshot). A family that is born and has no process noise reads `Q none` in that line.

Table 1: The dynamic state families of `StateManager` (`kStateFamilies`). The configuration keys of the process noise are under `filter.process_noise`, those of the initial standard deviation under `filter.initial_std`.

Family	Key example	Born under	Process noise	Initial std
slant ionosphere	`G05`	`processing.uncombined_enabled`, not a baseline mode	obliquity random walk, `ionosphere_q`	`ionosphere`
ionosphere rate	`G05`	and `estimation.enable_iono_rate`	CWNA with the level, `iono_rate_psd`	`ionosphere_rate`
ambiguity	`G05_L1C`	every processed phase row	random walk, `ambiguity_q`	1000 m, no key
receiver code bias	`GPS_IFB5`	uncombined and `estimation.estimate_ifb`	random walk, `code_bias_q`	`code_bias`
satellite code bias	`G05|C1W`	`estimation.estimate_sat_biases`	random walk, `sat_bias_q`	`sat_bias`
satellite phase bias	`G05|L1C|PB`	network, `estimation.estimate_sat_phase_biases`	random walk, `sat_phase_bias_q`	`sat_phase_bias`
receiver phase bias	`GPS_RPB_L1C`	network, `network.phase_datum_eq34`	none	`sat_phase_bias`
phase colour	`G05`	`network.phase_colour_state`	none	set by the creating code
satellite clock	`G05|SC`	network, `estimation.estimate_sat_clocks`	random walk, `sat_clock_q`	`sat_clock`

The rows with a network birth condition name parameters that a network run keeps in the station's own parameter set, not in `StateManager` (Section 2.5). The core block (position, clock, ZWD, velocity, gradients, inertial states, inter-system offsets) is index-stable by construction and is not a family; its process noise is built by `StateManagerProcessNoise.cpp`.

2.4 Dynamic indexing

`StateManager` addresses the variable-dimension blocks through key→index maps. `include/filter/StateKeySchema.hpp` defines every key format once, each with a `make` builder and, for the categories that are read back, a `parse` helper (the ionosphere and inter-system time-offset keys have no parser, since the key is the bare satellite or system name). Some call sites still build keys by string concatenation rather than through the builders (for example the double-difference keys in `MeasurementModel`, `DdPivotManager` and `KalmanFilter`); the strings they produce match the schema formats. The one function that renders a typed parameter identity as a user-side key is `userKey` (`measurement/UserParamView.hpp`). The formats are:

- iono key: `"<sys><prn>"`, e.g. `"G05"` (delimiter-free);
- ambiguity key: `"<sys><prn>_<signal>"`, e.g. `"G05_L1C"`, split on the *first* underscore, so an iono-free combination `"G05_L1C_L5Q"` retains its compound signal;
- satellite signal-bias key (code): `"<sys><prn>|<signal>"`, e.g. `"G05|C1W"`, with a pipe delimiter and its own map, disjoint from the underscore ambiguity namespace;
- satellite phase-bias key (network): `"<sys><prn>|<signal>|PB"`, e.g. `"G05|L1C|PB"`, where the trailing `|PB` keeps it disjoint from the code-bias namespace;

- receiver phase-bias key (network): `"<system>_RPB_<signal>"`, e.g. `"Galileo_RPB_L5Q"`, where the `_RPB_` infix keeps it disjoint from the receiver code-bias namespace (which matches on `IFB`/`DCB`);
- satellite clock key (network): `"<sat>|SC"`, e.g. `"G05|SC"`, per satellite rather than per signal; the trailing `|SC` discriminator gives the clock its own namespace;
- receiver code-bias key: `"<system>_IFB<band>"` or `"<system>_DCB"`; the per-slot GLONASS variant appends an FCN marker, `"GLONASS_IFB<band>_k<±slot>"` (e.g. `"GLONASS_IFB1_k-4"`);
- receiver inter-signal code-bias key (network): `"<system>_ISC_<signal>"`, e.g. `"GPS_ISC_C1W"`, stored in the receiver code-bias map; the `_ISC_` infix matches neither the IFB nor the DCB parser;
- inter-system time-offset key: the bare system name (`"Galileo"`, ...; GPS is the reference and never appears);
- double-difference ambiguity key (baseline): `"DD|<cons>|<sat>|<code>|<pivot>"`.

Addressing states by key rather than by position removes the index bookkeeping errors of hand-managed dynamic state vectors, and lets the BSSD and DD S-transforms (Section 4) operate as block operations on the relevant key sets.

Two further categories in the schema header are absent from the list above, because they are not rows of $\boldsymbol{x}$. The first is the BSSD pivot absolute-seed lookup key (`"<cons>|<band>"`). Its builder has no production caller and it has no parser: `MeasurementModel` builds the `pivot_abs_seeds` keys by concatenation, with the full phase signal code in place of the band (`"GPS|L1C"`), plus narrow-lane and wide-lane variants under the `nl_wl` basis. The second is the spherical-harmonic VTEC coefficient key (`"SH|<n>|<m>|<C|S>"`, with the second shell on the distinct `SH2` prefix), which names a column of the network's shared border, not a `StateManager` state, and has a `make`/`parse` pair. It belongs to the optional co-estimated map that the reference configurations do not use. Both appear in the generated key-format table of the appendix, which is derived from the schema header by a script and is therefore the completeness reference for the list above.

Ambiguity basis. By default the ambiguity block carries one per-band float ambiguity per tracked (satellite, signal), in metres. Setting `estimation.state_ambiguity_basis = "nl_wl"` switches the whole run to a narrow-lane / wide-lane parameterisation. Each satellite has a reference band, the lowest-numbered band it is observed on that epoch, and carries one narrow-lane state tied to that band (key `"<sat>_NL_<band>"`) and one wide-lane state for each other observed band, paired with the reference band (`"<sat>_WL_<a>_<b>"`, with the band digits in ascending order, semantically $N_a - N_b$). A change of reference band produces new keys, and the old states are pruned. Both kinds are in cycles rather than metres, so the wide-lane and LAMBDA arithmetic stays native to integers; the Jacobian carries the wavelength scaling (Table 2). The two bases never coexist in a single state vector. The network estimator does not use this basis.

2.5 Network station parameters

The network estimator runs no `KalmanFilter` and keeps no `StateManager`. Each station owns a parameter set (`StationParams`, `network/state/StationParams.hpp`): its columns in column order, each with a typed identity (kind, form, satellite, signal), the linearisation point the epoch's rows are evaluated at, and the disposition the station split gave it. The set is the station's own state and the network its only writer. A station's layout holds the receiver position (held at the roster coordinate and never estimated), the receiver clock, the zenith wet delay and one inter-system bias per active non-reference constellation. The families born during the run are the per-satellite slant ionosphere, the ambiguities (always one per band), the receiver code biases and, when armed,

the phase colour. The set also names the shared satellite parameters the station's rows reference: the satellite clock correction and the satellite phase and code biases. The optional columns of the single-receiver filter (velocity, clock rate, phase clock, gradients, inertial states, NL/WL ambiguities, between-satellite differencing) are not part of it, and a network run that enables one is warned at start. The rows of a station are built from the set by the shared observation equation (Section 3.4); the births of the local families and their first-epoch seeds are in `network/assembly/LocalBirth.cpp` (the first-epoch seeding section of the estimator-infrastructure document).

Parameters as armed. The network declares each kind of parameter once (`network/assembly/ParamRegistry.hpp`) and prints a table at run start, `network/assembly/ParamInventory.hpp`, that counts every class of parameter (kind, constellation, signal or band) by what the network did with each instance on the epoch: estimated in the shared border, estimated in the station's local block, held at its seed outside the solve, absorbed (a shared parameter that a gate withheld from the station, another column taking it up), pinned (the station coordinates), dropped, or unobserved. A declared rule whose premise does not hold on the run is printed as a warning under the table.

Time linking. The estimator solves the normal equations of an epoch. It links epochs with explicit carried priors in information form, of effective variance $\mathrm{var}_{\mathrm{eff}} = \mathrm{var} + q\,\Delta t$, not through a $\boldsymbol{Q}(\Delta t)$ (the network-mode document). The satellite-side densities `sat_clock_q`, `sat_phase_bias_q` and `sat_code_bias_q` are read as those carry rates. They can be set through the legacy `filter.process_noise.*` keys or through the network-scoped `network.process_noise` class table, whose per-class fallbacks equal the legacy defaults.

Satellite clocks $c\,\Delta\delta t^i$. Gated by `estimation.estimate_sat_clocks`, which the network-estimator loader forces on together with the phase-bias flag (`GnssConfigLoadNetwork.cpp`, the line that logs *forcing estimate_sat_clocks + estimate_sat_phase_biases*) and which `GnssConfig::applyNetworkConsumerCouplings` turns off again for the closed-loop user, who takes the same quantity as a known correction. The network estimates an explicit per-satellite clock correction $\Delta\delta t^i$ (in metres, relative to the a-priori precise or real-time product clock already removed from the modelled value), so the emitted clocks stay consistent with the chosen S-system datum. There is one parameter per satellite, shared by all its signals. The satellite orbit stays fixed from the product; only the clock is estimated. It enters each row with the partial -1 (Section 3.4); its identity is spelled `"<sat>|SC"`.

Satellite phase biases $\bar{b}_s^i$. Gated by `estimation.estimate_sat_phase_biases`; the JSON default is off and the network-estimator loader forces it on, as above. The user side imports the satellite phase bias as a known SSR correction (the SSR-corrections document) or absorbs it into the float ambiguity. The network instead carries an explicit per-(satellite, signal) phase-bias parameter on the undifferenced phase rows: the estimable bias of Odijk [3], which lumps the network pivot-receiver ambiguity. On a single station's row it is collinear with the per-receiver float ambiguity; only the joint network estimator separates it, across stations (the explicit phase-bias section of the network-mode document). Its identity is spelled `"<sat>|<signal>|PB"`, disjoint from the code-bias namespace. It is near-stationary between epochs, with the carry rate `sat_phase_bias_q`.

Satellite code biases. In the network's satellite-OSB mode (`network.estimate_sat_code_bias`, armed in the reference configurations) the parameter set covers every raw code (i, s), not only the pairs the BIA file leaves uncovered, and the external `.BIA` is not used, so the network's own estimate

is the single code-bias source. The datum that makes it estimable is derived in the network-mode document. The carry rate is `sat_code_bias_q`.

Receiver code biases and inter-signal biases. The receiver code-bias rule is the one of the single-receiver filter (above). On the network, a code that is not the station's latched canonical code on its band gets the separate receiver inter-signal code bias. These columns have no Kalman time update, because the network runs no `KalmanFilter`; they are re-estimated each epoch unless `network.carry_rx_code_bias` is armed (loader default off; armed in the reference configurations). That setting adds a diagonal random-walk carry (var $+ q\,\Delta t$) held in the network run state. Two guards protect it: at load, the loader forces the carry off unless the code datum is time-linked (`network.code_datum_time_linked`); at inject time, an innovation gate (`rxCbInjectGate`, `network/driver/RunTypes.hpp`) withholds the fold when the carried level and the station's own estimate disagree at receiver-reset scale. With the satellite code biases estimated, the receiver inter-frequency biases of the bands outside a constellation's clock-reference pair stay in the set at the value their first-epoch seed gave them and are left out of the station's local block, unless `network.estimate_rx_extra_band_code_delay` is armed (the constellation-scope section of the network-mode document).

Receiver phase biases $\delta_{r,s}$. The receiver-side counterpart of the satellite phase bias, gated by `network.phase_datum_eq34` (the single-receiver-chain phase datum; see the network-mode document). The setting is off in the reference configurations, which use the held phase-datum basis of the float vehicle (the float-vehicle document). When armed, each $\delta_{r,s}$ is an estimable parameter, so the receiver fractional phase bias no longer lumps into the emitted satellite phase bias. It is per (receiver, system, signal), spelled `"<system>_RPB_<signal>"`. Unlike every other bias parameter described here, these are local, each observed by a single station. At `network.phase_datum_eq34_rpb_q_m2s` $= 0$ each $\delta_{r,s}$ is re-estimated every epoch; at the non-zero default ($3 \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$) a local time-linking prior toward the previous estimate, of variance var $+ q\,\Delta t$, suppresses the per-epoch bounce that a weakly observable local parameter otherwise injects, anti-correlated, into every individual float ambiguity. The prior is erased for any (i, s) whose chain frame stepped at that epoch's election (`network/solve/ChainDatum.hpp`).

Phase colour. With `network.phase_colour_state` a per-satellite nuisance state with the partial $+1$ on every phase row of the satellite is born at value zero; it is off in the reference configurations.

3 Observation equations

This section gives the *undifferenced* observation model about which the filter linearises at each epoch. The PPP-RTK parameterisation is uncombined: the filter consumes per-signal pseudorange and carrier phase directly and forms no ionosphere-free or wide-lane combination as a preprocessing step. The combinations (IF, MW, GF) are computed for diagnostics and slip detection only (the cycle-slip document; `slip/MwSlipDetector`, `slip/GraphicSlipDetector`) and do not enter the measurement update. The loader (`GnssConfig::loadProcessingSection_`) forces `uncombined_enabled=true` for the whole PPP-RTK family (the single-receiver PPP-RTK modes and the network estimator) and for the baseline double-difference modes, which process each signal independently (Section 4.3). Plain PPP mode keeps the flag at its default `false` and builds ionosphere-free combined observables (keys such as `C1W_C5X`); these enter the same observation equation (Section 3.4) and the update. The rest of this section describes the uncombined case.

3.1 Code (pseudorange) observation

For receiver r tracking satellite i on signal s, the pseudorange is

$$\boxed{P^i_{r,s} \;=\; \rho^i_r \;+\; c(\mathrm{d}t_r - \mathrm{d}t^i) \;+\; c\,\delta t_r^{(g(i))} \;+\; T^i_r \;+\; \mu_s\, I^i_{r,1} \;+\; B_{r,s} - B^i_s \;+\; \varepsilon^P_{r,i,s}.} \tag{4}$$

3.2 Carrier-phase observation

The carrier-phase observation, scaled to metres by λ_s, is

$$\boxed{\Phi^i_{r,s} \;=\; \rho^i_r \;+\; c(\mathrm{d}t_r - \mathrm{d}t^i) \;+\; c\,\delta t_r^{(g(i))} \;+\; T^i_r \;-\; \mu_s\, I^i_{r,1} \;+\; \lambda_s\, N^i_{r,s} \;+\; b_{r,s} - b^i_s \;+\; \varepsilon^\Phi_{r,i,s}.} \tag{5}$$

The two equations differ structurally in the sign of the ionospheric term: $+\mu_s I$ on code, $-\mu_s I$ on phase. Further remarks:

1. The phase ambiguity $N^i_{r,s}$ is an integer in principle, but the receiver and satellite phase biases $b_{r,s} - b^i_s$ contaminate it. A *float* solution absorbs these biases into $N^i_{r,s}$, which leaves a non-integer state with no structural advantage over the code observation. Integer-valued ambiguities therefore require either external Fractional Cycle Bias (FCB) products or differencing (Section 4), which cancels the common-mode bias terms.
2. The clock product $c\,\mathrm{d}t^i$ comes from the analysis-centre CLK file (post-processing) or from the broadcast clock refined by a streamed SSR correction (real time, the SSR-corrections document). The filter treats it as deterministic. Its error enters the observation: the part common to all satellites is absorbed by the receiver clock state, and the satellite-specific part remains in the residuals. The network estimator adds a per-satellite clock correction to the product clock as a parameter of its own (Section 3.4).
3. The system time offset $c\,\delta t_r^{(g(i))}$ is zero by construction when $g(i)$ is the reference constellation, which is GPS (the state manager keys no offset for GPS). With the receiver clock $c\,\mathrm{d}t_r$, a multi-constellation run therefore carries $|\mathcal{G}|$ clock-like states in total.
4. The slant tropospheric delay is the hydrostatic zenith delay mapped by the dry mapping function plus the wet zenith delay mapped by the wet mapping function (`PppObservation` fields `mf_d` and `mf_w`).
5. The equations omit the deterministic model corrections applied to the computed range: satellite and receiver antenna phase-centre offsets and variations, the relativistic terms, site displacement, and, on phase rows only, the carrier-phase wind-up. The wind-up follows the `nominal-wu93` sign convention.

3.3 Geometric range and linearisation

The geometric range $\rho^i_r = \|\boldsymbol{r}^i - \boldsymbol{r}_r\|_2$ enters Equations (4) and (5) non-linearly through the receiver position $\boldsymbol{r}_r$. The filter linearises about the current estimate $\hat{\boldsymbol{r}}_r$:

$$\rho^i_r \;=\; \hat{\rho}^i_r \;-\; (\boldsymbol{e}^i_{\hat{r}})^{\mathrm{T}}(\boldsymbol{r}_r - \hat{\boldsymbol{r}}_r) \;+\; \mathcal{O}\big(\|\boldsymbol{r}_r - \hat{\boldsymbol{r}}_r\|^2\big)\,, \tag{6}$$

with the unit line-of-sight

$$\boldsymbol{e}^i_{\hat{r}} \;:=\; \frac{\boldsymbol{r}^i - \hat{\boldsymbol{r}}_r}{\|\boldsymbol{r}^i - \hat{\boldsymbol{r}}_r\|_2}. \tag{7}$$

The position columns of the Jacobian row $\boldsymbol{H}_{(i,s)}$ are therefore $-(\boldsymbol{e}^i_{\hat{r}})^{\mathrm{T}}$. One linearisation per epoch suffices for sub-metre residual targets. For large initial position offsets the engine can relinearise within the epoch (`processing.relinearize_geometry`, the iterated-update section of the filtering document). The flag defaults to off in code; the PPP, PPP-RTK, baseline and real-time configurations in `config/` enable it, and the network reference configurations (`config/golden/golden_field_refound.json`, `config/golden/golden_sim_refound.json`) leave it off.

The satellite position $\boldsymbol{r}^i$ is a function of the transmission time. `Geometry::computeEmissionState` obtains the emission time in one step from the *pseudorange* rather than by iterating against the geometric range, which depends on $\boldsymbol{r}_r$. The time of flight $\tau = P/c$ and the satellite clock are subtracted from the receiver epoch. The receiver clock bias $\mathrm{d}t_r$ enters the receiver epoch as $+\mathrm{d}t_r$ and the pseudorange as $+c\,\mathrm{d}t_r$, so it cancels exactly. The remaining error $(I + T + \varepsilon)/c \approx 30$–$60$ ns displaces the satellite by less than a millimetre, below any PPP threshold. The seed pseudorange is the ionosphere-free combination of the clock-reference signals when available (`Geometry::extractPseudorange`), so the ionospheric part of that error applies only to the single-frequency fallback. A range-based fixed point is avoided on purpose: iterating against ρ^i_r would reintroduce an uncancelled $\mathrm{d}t_r$ and drive a positive-feedback loop through the receiver-clock state.

The one second pass is the `model_corrections.refine_emission_time` block in `Geometry::computeSatelliteGeometry` (default on). It re-runs `computeEmissionState` on a cleaned pseudorange: the seed minus the slant hydrostatic delay and the IF-combined ATX projection, plus the periodic relativistic clock term. This refines the pseudorange; it is not a range-versus-position iteration.

3.4 One equation for every parameter set

Equations (4) and (5) are written once in code, as the list of terms that `observationEquation` (`measurement/ObservationEquation.hpp`) returns for a row. Three callers build a row through it: the single-receiver filter (`MeasurementModel`), the network estimator's station rows (`networkStationRows`, `network/assembly/StationRows.cpp`) and the first-epoch seeds (`measurement/Seeds.cpp`, the first-epoch seeding section of the estimator-infrastructure document). The user's rows and the network's rows are therefore one equation evaluated over two different parameter sets. The predicted observable of a row is

$$\hat{z}^i_{r,s} \;=\; z^{i,\mathrm{mod}}_{r,s} \;+\; \sum_e \; a_e \sum_k h_{e,k}\, x_{e,k}, \tag{8}$$

where z^{mod} is the a-priori model value of the signal (the range at the linearisation position, the a-priori troposphere, the product clock and the deterministic corrections of the last remark above), e runs over the terms of Table 2 in the order listed, a_e is the term's own factor (one unless stated), $h_{e,k}$ the partial of the k-th parameter of the term and $x_{e,k}$ its value at the linearisation point. The innovation of the row is $\boldsymbol{\nu} = z - \hat{z}$ and the Jacobian row holds $a_e h_{e,k}$ in the column the parameter occupies. A term whose parameter does not exist in the parameter set is absent, which is the same as adding zero; a phase row whose ambiguity does not exist has no equation.

Parameter views. The caller owns the parameters. It passes a lookup that answers, for one typed identity (kind, form, satellite, signal), the column and the linearisation value, or no column when the parameter does not exist. The single-receiver filter answers from its state vector through `UserParamView` (`measurement/UserParamView.hpp`), which maps an identity to the key formats of Section 2.4 and to the fixed core columns. The network answers from the station's own parameter set through `NetView` (Section 2.5). A shared satellite parameter (clock, phase bias, code bias)

Table 2: Terms of the predicted observable, in the order of the sum (`ObsEffect` in `measurement/ObservationEquation.hpp`). μ_s is Equation (1).

Term	Parameter	Partial	Present on
range correction	receiver position $\boldsymbol{r}_r$ (three columns)	$-\boldsymbol{e}^i$, applied to $\boldsymbol{r}_r - \boldsymbol{r}_0$ (Equation (6))	every row
receiver clock	$c\,\mathrm{d}t_r$	$+1$	every row
phase clock	$c\,\mathrm{d}t_r^{\phi}$	$+1$	phase rows
troposphere	ZWD_r	$m_w(\theta_r^i)$	every row
gradients	(g_N, g_E)	$m_g(\theta)\cos\alpha$, $m_g(\theta)\sin\alpha$ (Equation (3))	every row
inter-system bias	$c\,\delta t_r^{(g)}$, g not the time reference	$+1$	every row
slant ionosphere	$I_{r,1}^i$	$+\mu_s$ on code, $-\mu_s$ on phase	uncombined rows
receiver code bias	$B_{r,\cdot}$ (Section 2)	coefficient of the receiver-bias rule (1, or the geometry-free coefficient of the code-bias seed of the estimator-infrastructure document)	code rows, uncombined, with `estimation.estimate_ifb`
satellite code bias	B_s^i	$+1$	code rows, with `estimation.estimate_sat_biases`
satellite phase bias	$\bar{b}_s^i$ and, with the receiver phase bias, $\delta_{r,s}$	$+1$ each	uncombined phase rows, with `estimation.estimate_sat_phase_biases`
satellite clock	$\Delta\delta t^i$	-1	uncombined rows, with `estimation.estimate_sat_clocks`
SSR bias	none (a known value)	the decoded bias times $+1$	every row; none on the network's own rows
ambiguity	λN in metres; on the NL/WL basis N_{NL} and N_{WL} in cycles	$+1$; on the NL/WL basis λ_s and $\mp\lambda_s$	phase rows
phase colour	per-satellite nuisance	$+1$	phase rows, with `network.phase_colour_state`

answers with its border column and the value zero, because it is absolute in the joint solve and is never written back to a station. A seed asks for fewer terms: it subtracts only those known when its state is born, selected by a mask of terms (the first-epoch seeding section). A network station always carries one ambiguity per band, whatever `estimation.state_ambiguity_basis` says.

Signs shared by the network and the user. The network estimates the satellite biases and clocks as parameters; a user of the network's corrections takes the same quantities as known terms. The two agree only if the partial of the estimated parameter and the multiplier of the known term are the same number, so one constant each serves both sides: $+1$ for a satellite bias, code or phase (`kSatBiasSign`), and -1 for the satellite clock correction (`kSatClockPartial`). The clock enters with -1 because the observable carries $-c\,\mathrm{d}t^i$ and the a-priori product clock is already inside the modelled value. The imported bias sign is treated with the SSR mapping (the SSR-corrections document). Doppler rows are not part of this equation: velocity, clock rate and the inertial states are columns only a single-receiver configuration has.

3.5 Stochastic model

The observation noise covariance of an epoch is diagonal, with entries

$$\sigma^2_{r,i,s} \;=\; \sigma^2_{0,\mathrm{type}(s)} \;\cdot\; w(\theta^i_r)^{-1} \;\cdot\; \xi(\mathrm{C/N_0 \text{ at } } (r,i,s)), \tag{9}$$

The three factors are:

- a per-type baseline ($\sigma_{0,\mathrm{code}}$, $\sigma_{0,\mathrm{phase}}$ from `filter.measurement_noise`);
- an elevation down-weighting factor $w(\theta^i_r)^{-1} = (\sin\theta^i_r)^{-2p_{\mathrm{type}}}$ with a continuous, type-dependent exponent. Code and Doppler rows use `filter.code_elevation_power` and carrier-phase rows `filter.phase_elevation_power`; both default to 1.0, which scales the variance by the squared cosecant of the elevation. When the sine of the elevation is at most `1e-6`, a guard returns σ_0^2 unamplified. The derivation is in the stochastic-model section of the quality-control document;
- a carrier-to-noise-ratio term ξ, parameterised by `snr_model_params`.

`filter.stochastic_model` selects the weighting strategy: `constant`, `elevation` (the default), `snr` or `satellite_block`. Only the `snr` strategy uses the carrier-to-noise term, and it does not multiply the baseline variance by a factor: for code and phase rows it replaces the baseline variance by the variance-additive Hadas–Bosy form built from the observed carrier-to-noise ratio (coefficients `code_a`, `code_b`, `phase_a`, `phase_b`; the ratio is clamped to 20–55 dB-Hz by default; code rows take a per-band scale), then applies the elevation factor (`StochasticModelWeight::computeSnr`). Cross-correlations between observations are zero in the undifferenced parameterisation; differencing introduces the structured correlations treated in Section 4.

The variance of a row is Equation (9) on the network estimator's own rows (`rowVariance` in `measurement/MeasurementModelVariance.cpp`). A user of the network's corrections adds the uncertainty of the corrections it applies to the same variance: the satellite accuracy index with its clock-staleness growth, the phase-bias sigma on phase rows, the combined-correction sigma where it replaces those two, and a down-weighting factor on code rows without a bias-product entry (the SSR-corrections document).

4 Differencing transformations

PPP-RTK with the single-difference transform and the baseline modes do not consume the undifferenced observations of Equations (4) and (5) directly. Each forms a linear combination across satellites (BSSD), receivers (BRSD), or both (DD). The combination cancels common-mode terms and leaves a residual structure in which the ambiguities can be integer. This section gives the algebraic cancellations and the matrix-form S-transforms that map undifferenced parameters to differenced ones.

4.1 Between-satellite single difference (BSSD)

Each constellation g has a pivot satellite p_g, elected by `PivotManager::selectPivotCandidates` under a sticky retention policy (the ordering is given in the *Pivot selection* paragraph below). The same `PivotManager` pivot serves both the combined (ionosphere-free) and the uncombined parameterisation. For any non-pivot satellite $i \in \mathcal{S}_g \setminus \{p_g\}$ and signal s, define

$$\begin{aligned} \Delta P_{r,s}^{i,p_g} &:= P_{r,s}^{i} - P_{r,s}^{p_g}, \\ \Delta \Phi_{r,s}^{i,p_g} &:= \Phi_{r,s}^{i} - \Phi_{r,s}^{p_g}. \end{aligned} \tag{10}$$

Substituting Equations (4) and (5) and cancelling the receiver-side common terms ($c\,\mathrm{d}t_r$, $c\,\delta t_r^{(g)}$, $B_{r,s}$, $b_{r,s}$) yields

$$\Delta P_{r,s}^{i,p_g} = \Delta\rho_r^{i,p_g} - c\,\Delta\mathrm{d}t^{i,p_g} + \Delta T_r^{i,p_g} + \mu_s\,\Delta I_{r,1}^{i,p_g} - \Delta B_s^{i,p_g} + \Delta\varepsilon^{P}, \tag{11}$$

$$\Delta \Phi_{r,s}^{i,p_g} = \Delta\rho_r^{i,p_g} - c\,\Delta\mathrm{d}t^{i,p_g} + \Delta T_r^{i,p_g} - \mu_s\,\Delta I_{r,1}^{i,p_g} + \lambda_s\,\Delta N_{r,s}^{i,p_g} - \Delta b_s^{i,p_g} + \Delta\varepsilon^{\Phi}. \tag{12}$$

The receiver-side cancellations are exact only when both observations enter the filter with the same modelling (tropospheric mapping, antenna PCV correction and so on). The satellite-dependent terms do not cancel: ΔT, for example, is the difference of two slant delays at different elevations and is small but not zero.

`BssdTransform::apply` realises the cancellation by zeroing the design-matrix columns of the receiver-shared states after differencing: receiver clock, phase clock, ISB, clock rate and ISB rate (when allocated), and the receiver code-bias (DCB/IFB) states. In both parameterisations it also zeroes the pivot's ambiguity columns (under the NL/WL basis, the pivot's NL column and, on a non-reference band, its WL column).

The cancellation of $B_{r,s}$ in Equation (11) holds for CDMA constellations, whose group delay is a single per-signal receiver term. GLONASS is the exception. Its FDMA inter-frequency biases are per-FCN-slot states, so two satellites on different slots carry different receiver delays and a residual $B_{r,s}^{(k(i))} - B_{r,s}^{(k(p_g))}$ survives the single difference. `BssdTransform` therefore exempts the per-slot GLONASS IFB states (identified by `StateKeys::isGlonassPerSlotIfbKey`) from the column zeroing. `PivotManager::buildEpochMaps` drops GLONASS from the BSSD candidate maps unless the flag is set, and `BssdTransform::apply` drops any row whose group has no pivot, so with the flag off GLONASS contributes no single-differenced row. GLONASS BSSD is off by default (`estimation.between_satellite_sd.enable_glonass`); when enabled, Equation (11) acquires the surviving inter-slot IFB difference as an additional estimated column.

Pivot selection. The incumbent pivot is kept while it remains at or above `estimation.between_satellite_sd.pivot_min_elevation` (default 45 degrees), is not in the exclusion set, and passes the

ephemeris-freshness bar described below. Otherwise a new pivot is elected. The primary key is a hard lexicographic tier on phase-signal coverage: a full-coverage candidate outranks any partial-coverage candidate regardless of score, because the BSSD frame requires the pivot to track every signal another satellite tracks. Within a coverage tier the additive score $-\min\bigl(1, \sqrt{\sigma^2_{\hat{N},\max}}\bigr) + \mathrm{el}/90 - \frac{1}{2}\mathbb{1}[\mathrm{el} < 20^\circ]$ ranks the candidates. Float-ambiguity convergence $\sqrt{\sigma^2_{\hat{N},\max}}$ and elevation enter as co-equal terms, with a low-elevation penalty below 20°; a candidate with no ambiguity state yet takes the saturated convergence penalty of 1. The ephemeris-freshness bar excludes from pivot election any candidate whose broadcast $|\mathrm{obs} - \mathrm{toe}|$ exceeds that of its constellation's freshest qualified peer by more than `kPivotToeSpreadMaxS` (1200 s); the satellite remains usable as a non-pivot. The bar prevents the extrapolated orbit and clock error of a stale-ephemeris pivot from entering every SD row it anchors. Candidates below the minimum elevation are excluded; only when no satellite of the constellation passes the filters does the election fall back to the best-ranked or highest satellite without them. The baseline DD path scores differently: `DdPivotManager` keeps the incumbent while it remains in the candidate set of satellites shared by both receivers, and otherwise ranks on $\mathrm{avg_snr} + 0.01 \cdot \mathrm{num_sigs}$, without elevation.

Pivot transitions. A handover has three kinds of trigger. The first is selection: when the incumbent fails the retention test above (it leaves the visible set, drops below the minimum elevation, is excluded, or fails the freshness bar), p_g is re-elected from the remaining satellites. The second is a disturbed state: `isPivotDisturbed` reports the pivot disturbed when any of its phase ambiguity states has left $(0, 0)$ (absolute value above 1×10^{-6} or variance above 1×10^{-12}), for example after a cycle-slip reset. The third is an event: the Kalman filter's post-fit residual analyser (the pivot-offset detectors of the gating section of the filtering document) calls `PivotManager::requestForcedHandover` when the pivot's observation has slipped while its filter state is intact, a case `isPivotDisturbed` cannot detect. The forced request carries an exclusion set, so the handover cannot land on another offset satellite. The `retired_pivots_this_epoch_` set bars a pivot retired in the current epoch from re-election within it, which terminates handover cascades, and the same-epoch re-entry gate (`hasPendingForcedHandover`) applies the handover before the rebuilt measurement update. The request also carries a re-inflation set: non-pivot states that absorbed a coherent pivot-side code bias have their covariance re-inflated after the re-reference (when slip-state re-inflation is enabled), because the information-conserving transform below cannot remove that offset. The baseline DD path has the counterpart `DdPivotManager::requestForcedRerank`, which drops the incumbent and excludes the flagged satellites from the next re-rank. The event response is required because the pivot's states are frozen at zero variance (below): a stiff, time-linked zero-q quantity requires a wired jump response.

On a handover the float ambiguities $\{\Delta N_{r,s}^{i,p_g^{\mathrm{old}}}\}_{i,s}$ are re-expressed relative to p_g^{new} by the linear S-transformation $\boldsymbol{x}^{\mathrm{new}} = \boldsymbol{S}\,\boldsymbol{x}^{\mathrm{old}}$, with $\boldsymbol{S}$ chosen so that $\Delta N_{r,s}^{p_g^{\mathrm{new}},p_g^{\mathrm{old}}}$ becomes the new reference (zero). The covariance transforms as $\boldsymbol{P}^{\mathrm{new}} = \boldsymbol{S}\,\boldsymbol{P}^{\mathrm{old}}\,\boldsymbol{S}^{\mathrm{T}}$. The transform conserves the filter's information and so preserves the integer information accumulated in the ambiguities. In the uncombined PPP-RTK parameterisation `PivotManager` applies the same $\boldsymbol{S}\,\boldsymbol{P}\,\boldsymbol{S}^{\mathrm{T}}$ re-reference to three state groups (`reReferenceAmbiguities`, `reReferenceIonosphere`, which also moves the ionosphere-rate states when they exist, and `reReferenceSatBiases`). This is why `BssdTransform` zeroes the pivot's ionosphere and sat-bias $\boldsymbol{H}$-columns after differencing, and why $\Delta I_{r,1}^{i,p_g}$ in Equations (11) and (12) is a pivot-referenced SD quantity rather than an absolute slant delay.

Where the transform cannot carry a state onto the new pivot, a re-seed follows or replaces it. In PPP-RTK three cases arise. An ambiguity on a phase signal the new pivot does not track stays

referenced to the old pivot, so it is reset and marked for re-seeding against the new pivot. When the old pivot was disturbed, its re-referenced ambiguities are re-seeded from the single-differenced observations. When a disturbed pivot has no eligible replacement (every other satellite of the constellation is excluded or absent), `PivotManager::handleDisturbedPivot` re-zeroes the pivot's phase ambiguity states in $\boldsymbol{x}$ and the corresponding rows and columns of $\boldsymbol{P}$ in place, and no handover takes place. In the baseline DD path `DdPivotManager` applies the transform per (cons, code) group; a group in which the new pivot has no existing ambiguity is skipped (logged as "reset-and-reseed"), and its stale rows are pruned and re-seeded on the new pivot.

The S-basis constraint is realised by freezing the pivot's ambiguity, ionosphere (and ionosphere-rate, when enabled) and sat-bias covariances to zero and suppressing their process noise (`freezePivotAmbiguityCovariance`, `freezePivotIonoAndBiasCovariance`). The frozen ambiguity state is the condition `isPivotDisturbed` tests.

`PivotManager::getCleanHandovers` reports the *clean* handovers of the epoch: those that were neither forced nor triggered by a disturbed pivot. A clean handover is a lossless, invertible frame change, so the RTS smoother inverse-references the affected states across it; at a forced or disturbed handover the old pivot's ambiguity is discontinuous and the smoother starts a new segment. The losslessness stated above holds only for the clean subset.

The two modes use distinct code paths that share the algebra above. In PPP-RTK the pivot election, the sticky retention policy and the re-reference live in `filter/PivotManager`. The baseline DD path performs the re-reference in `filter/DdPivotManager`, under the `baseline.enable_pivot_handover` flag (default on). There the S-transform is applied in place, as a row pass over the state vector and covariance followed by a column pass over the covariance, without forming the transform matrix; the result passes through the positive-semidefinite guard `CovReReference::enforcePsd` (whose only caller is `DdPivotManager`), and the ambiguity keys are renamed onto the new pivot. With the flag off, every pivot change, including a forced re-rank, erases the DD ambiguities and re-seeds them.

4.2 Between-receiver single difference (BRSD)

Baseline modes process one or more rovers r_{rov} against a single fixed-coordinate reference r_{ref}; each rover has its own measurement model. The BRSD operator forms

$$\nabla P^{i}_{r_{\mathrm{rov}} r_{\mathrm{ref}},s} \;:=\; P^{i}_{r_{\mathrm{rov}},s} - P^{i}_{r_{\mathrm{ref}},s}, \tag{13}$$

and analogously for $\nabla\Phi$. The cancellations of $\nabla\cdot$ are dual to those of $\Delta\cdot$: BRSD cancels the satellite-side common terms ($c\,\mathrm{d}t^{i}$, B^{i}_{s}, b^{i}_{s}) and retains the difference of the two stations' receiver clocks and biases. Once ∇ is applied, the satellite biases need no state in the filter. `BrsdTransform::apply` carries only the rover-side line of sight in its output: in baseline mode only the rover position is estimated, so the reference-station line of sight multiplies no state.

4.3 Double difference (DD)

The composition $\nabla\Delta\cdot = \Delta\big(\nabla\,\cdot\big)$ is order-independent up to noise correlation: $\Delta\big(\nabla\,\cdot\big) = \nabla\big(\Delta\,\cdot\big)$. Applied to the carrier-phase model, with the common-mode terms cancelled, it gives

$$\nabla\Delta\Phi^{i,p_g}_{r_{\mathrm{rov}} r_{\mathrm{ref}},s} \;=\; \nabla\Delta\rho^{i,p_g}_{r_{\mathrm{rov}} r_{\mathrm{ref}}} + \nabla\Delta T^{i,p_g}_{r_{\mathrm{rov}} r_{\mathrm{ref}}} - \mu_s\,\nabla\Delta I^{i,p_g}_{r_{\mathrm{rov}} r_{\mathrm{ref}},1} + \lambda_s\,\nabla\Delta N^{i,p_g}_{r_{\mathrm{rov}} r_{\mathrm{ref}},s} + \nabla\Delta\varepsilon^{\Phi}. \tag{14}$$

Equation (14) retains the geometry term (linear in the rover position perturbation), the differential atmospheric delays and the integer DD ambiguity. This structure is the basis of carrier-phase RTK.

Short vs. long baselines. On short baselines ($\lesssim 10\,\mathrm{km}$) the differential atmospheric terms in Equation (14) are negligible at the millimetre level and are not modelled. The differential ionosphere $\nabla\Delta I$ grows linearly with separation and reaches typically 1 m at a few hundred kilometres, which exceeds the carrier wavelength. `GnssProcessor` offers two mitigations, both off by default:

- `baseline.use_iono_free` collapses the per-band DD rows into the dual-frequency ionosphere-free combination. It eliminates the first-order ionospheric term at the cost of a non-integer ambiguity (IF phase ambiguities stay float and are excluded from AR) and $\sim 3\times$ amplified code noise. Satellite pairs with fewer than two shared bands are dropped in this mode.
- `baseline.estimate_differential_tropo` adds a single $\Delta\mathrm{ZWD}_{r_{\mathrm{rov}}r_{\mathrm{ref}}}$ random-walk state (`differential_tropo_init_std_m`, default 0.1 m; `differential_tropo_process_noise_m_per_sqrt_s`, default 1×10^{-4}). It absorbs the differential wet delay that survives the BRSD on long baselines or baselines with a large height difference.

4.4 S-transform structure

BSSD and DD both have a matrix form $\boldsymbol{x}_{\mathrm{diff}} = \boldsymbol{S}\,\boldsymbol{x}_{\mathrm{undiff}}$, with $\boldsymbol{S}$ a block design transform. For BSSD restricted to the ambiguity block of one constellation g on one signal s, with $|\mathcal{S}_g|$ tracked satellites, $\boldsymbol{S}$ is a $(|\mathcal{S}_g|-1) \times |\mathcal{S}_g|$ matrix of the form

$$\boldsymbol{S}_{\mathrm{BSSD}} = \begin{bmatrix} 1 & 0 & \cdots & 0 & -1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & -1 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots & & & \\ 0 & 0 & \cdots & 0 & -1 & 0 & \cdots & 1 \end{bmatrix}, \tag{15}$$

where the column of -1's sits at the index of the pivot p_g. The DD operator is the composition of the two single-difference operators, $\boldsymbol{S}_{\mathrm{DD}} = \boldsymbol{S}_{\mathrm{BSSD}}\,\boldsymbol{S}_{\mathrm{BRSD}}$, realised in code as `BrsdTransform::apply` followed by `BssdTransform::differenceBrsd` (`include/measurement/BssdTransform.hpp`). The composition coincides with the Kronecker product $\mathbf{D}_{\mathrm{sat}} \otimes \mathbf{D}_{\mathrm{rx}}$ of the between-satellite and between-receiver differencing matrices acting on the vectorised observations only on a fully populated satellite $\times$ receiver observation grid. The Kronecker form is a model; real data give a non-rectangular grid, and the code differences whatever pairs exist. The transformed covariance $\boldsymbol{P}_{\mathrm{diff}} = \boldsymbol{S}\,\boldsymbol{P}_{\mathrm{undiff}}\,\boldsymbol{S}^{\mathrm{T}}$ is in general dense even when $\boldsymbol{P}_{\mathrm{undiff}}$ is diagonal, because every differenced row shares the pivot column. A DD filter therefore cannot be implemented as independent per-pair scalar updates: the correlated joint update is what integer ambiguity recovery through the LAMBDA decorrelation requires. Teunissen [5] gives the general theory and a catalogue of admissible S-transforms; Jonge [2] treats the rank defect.

Implementation notes. `GnssProcessor` does not materialise the chosen S-system as an explicit minimum-constraint $\boldsymbol{S}$-basis matrix. It is realised implicitly by the pivot-differencing operator (`measurement/BssdTransform`), an S-basis-invariant realisation of the uncombined parameterisation of Odijk [3]. A `DifferencingMode` strategy in `measurement/MeasurementBuilder` dispatches the three regimes (undifferenced, BSSD and DD) to separate builder classes.

Integer recovery on a DD row requires all four contributing observations to share the same RINEX signal code. `BssdTransform::differenceBrsd` marks a row `integer_capable` only when three conjuncts hold: `!brsd_aliased` on both the i_i and the pivot BRSD entries (rover and reference agree on each, i.e. no cross-attribute alias), and the rover uses the same code on i_i as on the pivot ($\texttt{rov_code}_i = \texttt{rov_code}_{\mathrm{piv}}$). The third conjunct is independent of the `brsd_aliased` flags: a row

can be un-aliased on both entries and still carry a non-integer DCB/IFB residual if the rover mixes codes across i_i and the pivot. Rows that fail any conjunct are carried as float and excluded from ambiguity resolution.

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